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Advanced Modeling of GaN-on-Silicon Spiral Inductors

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Abstract

In this paper, the accuracy of basic and advanced spiral inductor models for gallium nitride (GaN) integrated inductors is evaluated. Specifically, the experimental measurements of geometrically scaled circular spiral inductors, fabricated in a radio frequency (RF) GaN-on silicon technology, are exploited to estimate the errors of two lumped geometrically scalable models, i.e., a simple π-model with seven components and an advanced model with thirteen components. The comparison is performed by using either the standard performance parameters, such as inductance (L), quality factor (Q-factor), and self-resonance frequency (SRF), or the two-port scattering parameters (S-parameters). The comparison reveals that despite a higher complexity, the developed advanced model achieves a significant reduction in SRF percentage errors in a wide range of geometrical parameters, while enabling an accurate estimation of two-port S-parameters. Indeed, the correct evaluation of both SRF and two-port S-parameters is crucial to exploit the model in an actual circuit design environment by properly setting the inductor geometrical parameters to optimize RF performance.

Keywords: Gallium nitride (GaN); on-wafer measurements; radio frequency (RF)-integrated circuits; S-parameters; spiral inductors

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1. Introduction

Spiral inductors are fundamental components in radio frequency (RF) integrated circuits (ICs) since they are essential in crucial functionalities, such as impedance matching, tuned resonant load, LC oscillator tanks, etc. Although inductor devices are usually available in most RF technologies, inductor modeling, which is required for effective RF IC design, is often not accurate enough, thus leading to an extensive use of electromagnetic (EM) simulations. In recent years, GaN-based High-Electron-Mobility Transistors (HEMTs) gained attention in high-power applications thanks to better voltage/current capability in a wide temperature operative range. However, very few works about inductor measurements and modeling have been published with regard to GaN technologies so far [1–6]. Specifically, an equivalent circuit extracted for a single inductor geometry was proposed in [5] while lumped scalable models were proposed only in [2,6]. While in silicon-based technologies, many inductor models with different topologies and fitting equations are available [7–10], GaN-based inductor modeling is still poor despite the increasing success of such technology, which could be a bottleneck in the future development of GaN RF ICs. Indeed, the lack of accurate geometrically scalable

models for integrated inductors extends project timelines, since time-consuming electromagnetic (EM) simulations must be used in the very first design phase. Recently, neural networks, machine learning, or deep learning techniques have demonstrated interesting results in the automatic design of mm-wave ICs and component modeling [11–18].

In this paper, two lumped scalable models are used to estimate the electrical parameters of geometrically scaled spiral inductors fabricated with RF GaN technology on silicon for sub 6 GHz power applications. Specifically, the first is a basic π-model, like the one in [6], previously developed from EM-based data in a similar GaN technology. The latter is based on the model developed in [10] for silicon-based spiral inductors, which is here modified for the adopted GaN-on-silicon process. The accuracy of both models is evaluated and compared in terms of either standard performance parameters, such as inductance (*L*), quality factor (*Q*-factor), and self-resonance frequency (*SRF*) or the two-port scattering parameters (*S*-parameters). The comparison has been carried out by using on-wafer measurements of a circular spiral in a wide range of inductances and geometric parameters to evaluate model prediction capabilities as a RF IC design tool.

In this paper, the adopted RF GaN-on-silicon technology is presented in Section 2 and the two models under comparison are detailed in Section 3. Section 4 reports the accuracy comparison between models and conclusions are drawn in Section 5.

2. RF GaN Technology

This study has been carried out in an RF GaN process fabricated on a thin silicon (Si) substrate. This technology has been developed by STMicroelectronics, mainly for sub 6-GHz band power applications [18,19]. It includes 50-V HEMTs with a gate length of 0.5-μm, optimized for normally on depletion operation, thus exploiting the total two-dimensional electron gas (2DEG) density at the AlGaN/GaN heterointerface. The technology also provides resistors and high-*Q*-factor metal–insulator–metal (MIM) capacitors for the integration of biasing and matching networks, respectively, while allowing for the fabrication of polygonal/circular spiral inductors and interleaved/stacked transformers that are essential for advanced RF ICs [20–26]. The BEOL of the process consists of two top thick AlCu metals (namely MTL3 and MTL2), used for the inductor coil and the underpass, respectively, and a thinner AlCu layer for transistor connections. Specifically, the thickness of MTL3 and MTL2 is equal, which allows inductor series resistance to be reduced while avoiding possible current limitations due to the underpass connection. The adopted GaN technology has been developed for watt-level sub 6-GHz power amplifiers (*PAs*) for base station applications. The aim is to replace the traditional micromodule approach based on discrete components with a fully integrated system, thus achieving a significant reduction in cost and size.

Figure 1 shows the microphotograph of a circular inductor in the adopted RF GaN technology and the SEM micrograph of the upper metal layers, as they are connected in the underpass region of a spiral inductor. Specifically, as shown in Figure 1a, the coil exploits the top MTL3, while the underpass is implemented in MTL2. The circular geometry has been exploited to achieve better *Q*-factor performance than squared or polygonal shapes. The ground substrate electrical connection is provided by the trough–silicon–vias (TSVs) that also guarantee proper thermal dissipation, especially in high-power applications. It is worth noting that Figure 1 depicts the spiral inductor in a two-port test structure with ground–signal–ground (GSG) input/output pads (i.e., DUT) used for on-wafer measurement. On-wafer measurements have been performed by means of a two-port VNA by Keysight [27] connected to a 200-nm Cascade summit prober [28]. LRM calibration has been adopted [29,30]. A 5-step de-embedding procedure was adopted to remove the RLC parasitic and refer the measurement plane to the inductor terminals,

highlighted in red in Figure 1a [31]. To this end, the de-embedding structures shown in Figure 2 were integrated and measured for each inductor geometry.

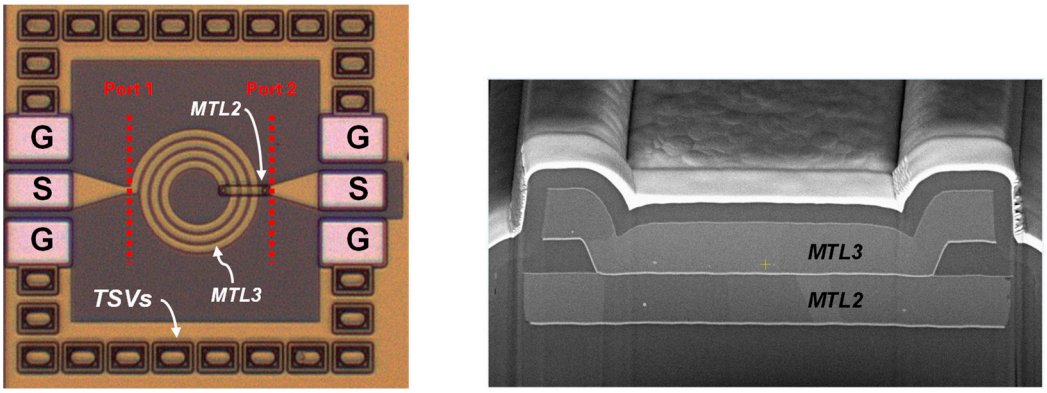


Figure 1. Microphotograph of a spiral inductor in the RF GaN-on-Si technology (**a**); SEM micrograph of the inductor in the underpass region (**b**) [6].

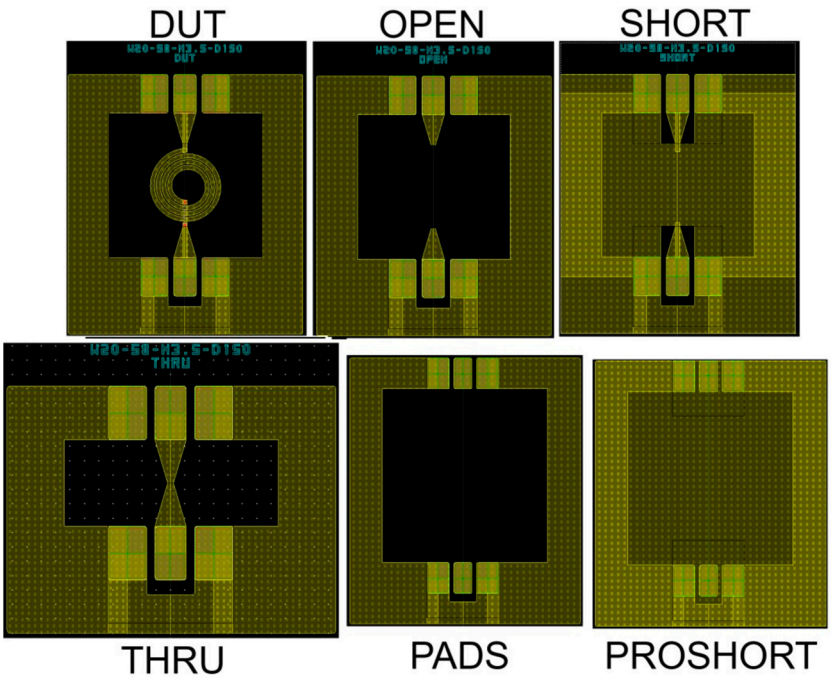


Figure 2. DUT and de-embedding structures used for inductor on-wafer measurement.

3. Inductor Lumped Model Descriptions

Inductor modeling is mainly aimed at predicting the electrical performance of integrated inductors in a large range of inductance values, which is essential to RF IC design. Indeed, an accurate lumped scalable model allows setting of the geometrical parameters of a spiral inductor, which are shown in Figure 3, to comply with RF circuit specifications (i.e., *L*, *Q*-factor, *SRF*, maximum current, consumed area). The availability of accurate scalable modeling reduces the use of time-consuming EM simulations in the very last design phase, according to the scheme sketched in Figure 4.

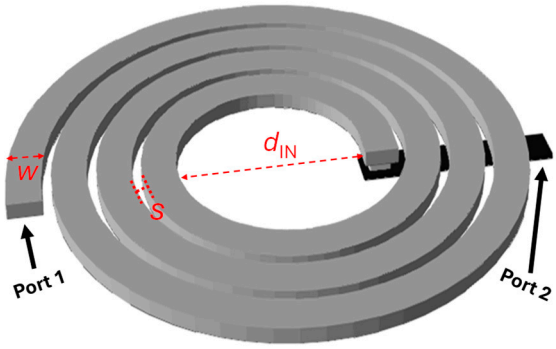


Figure 3. Geometrical parameters of a spiral RF inductor.

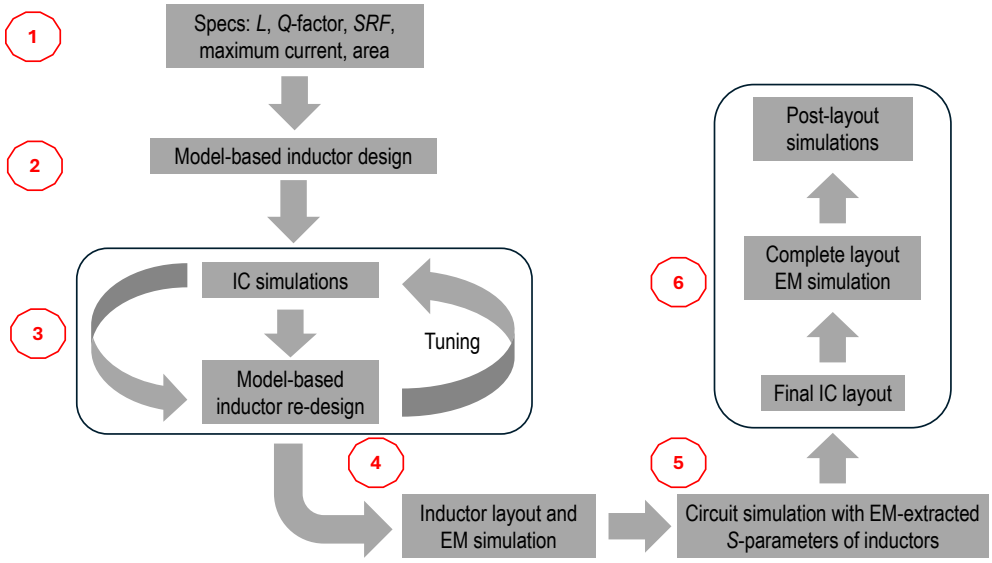


Figure 4. Simplified design flow of integrated inductors for RF ICs.

In this work, two different lumped models are proposed and compared in terms of accuracy. The models have been mainly developed to cover the typical needs of sub 6-GHz *PAs*. Their schematics are depicted in Figure 5. The first model shown in Figure 5a has been developed starting from the simple model in [6], making some improvements to increase its accuracy. The latter, shown in Figure 5b, is inspired by a topology proposed for silicon-integrated inductor modeling in [10]. In the following subsections, geometrically scaled equations for lumped components are detailed for both proposed models.

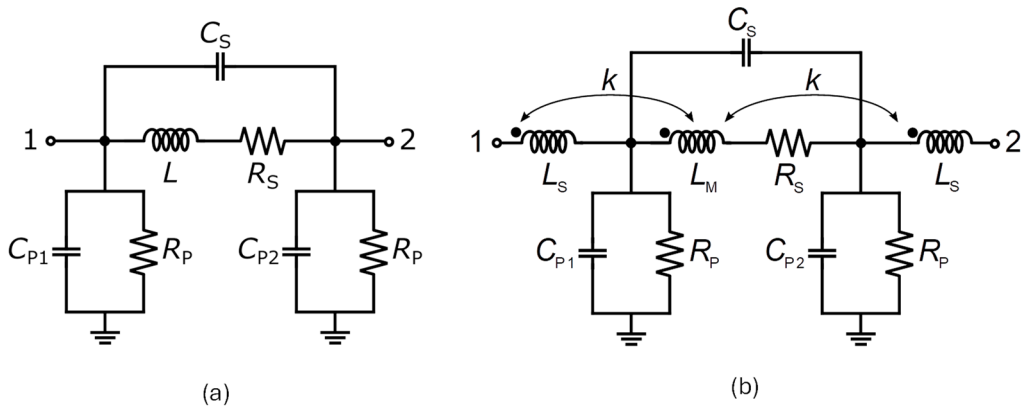


Figure 5. GaN-on-Si inductor models under comparison (Terminal 2 is the underpass of the inductor): (**a**) lumped π-model; (**b**) advanced lumped model**.**

3.1. Lumped π-Model

With reference to Figure 5a, the ideal inductor, L, models both spiral and underpass inductances, as well as the substrate magnetic coupling. Two different capacitors, C_{P1} and C_{P2}, account for the asymmetric substrate capacitive effects, while C_S models the fringing capacitances between the spiral turns, and the overlap capacitance between the spiral and the underpass. Both ohmic and substrate losses are estimated by means of frequency-variable resistances, R_S and R_P, respectively.

Specifically, L consists of three inductances,

$$L = L_{\mathrm{COIL}} + L_{\mathrm{UND}} + L_{\mathrm{SUB}} \tag{1}$$

where L_{COIL} and L_{UND} are the spiral and underpass inductances, respectively, while L_{SUB} accounts for the induced magnetic field in the substrate [32–34]. The first two terms of (1) are calculated by means of closed form expressions [35,36]. On the other hand, L_{SUB} adopts a similar monomial formula to [6], which better estimates the inductance reduction due to an induced magnetic field within the substrate:

$$L_{\mathrm{SUB}} = -20n^2 d_{\mathrm{AVG}}^2 w \tag{2}$$

where d_{AVG} is the average inductor diameter, n is the number of turns, and w the metal width.

The capacitor, C_S, is calculated by means of the same scalable equation used in [6]:

$$C_{\mathrm{S}} = \frac{1}{N_{\mathrm{OVER}}} \frac{\varepsilon_M w^2}{t_{\mathrm{M}}} p(n) \tag{3}$$

where N_{OVER} is the number of overlaps between the spiral and the underpass (i.e., $N_{OVER} = n + 1.5$), and ε_M and t_M are the electric permittivity and the thickness of the intermetal (i.e., MTL3-MTL2) dielectric layer, respectively. However, to improve the geometrical scalability of (3), a different polynomial expression function $p(n)$ is proposed in (4).

$$p(n) = n^2 - 3.3n + 3.8 \tag{4}$$

Substrate capacitances, $C_{P1,2}$, are calculated by means of the same equations reported in [6] that for the sake of completeness are reported hereinafter:

$$C_{\mathrm{P1,2}} = \frac{C_{\mathrm{SUB}} A_{\mathrm{COIL1,2}}}{2} \left[1 + (1 + \alpha_1) e^{-\beta_1 w}\right] \tag{5}$$

where C_{SUB} is the substrate capacitance per unit area, A_{COIL} is the coil area, α_1 is a fitting parameter equal to 0.92, and β_1 is an experimental coefficient estimated to be 3.53×10^4. The coil area, $A_{COIL1,2}$, is given by the product of the width, w, and the spiral length, $l_{COIL1,2}$ that are calculated according to the following expressions, where d_{IN} is the inner diameter:

$$l_{\mathrm{COIL1}} = \pi n [d_{\mathrm{IN}} + w + (n - 0.5)(w + s)] \tag{6}$$

$$l_{\mathrm{COIL2}} = \pi n \left[d_{\mathrm{IN}} + w + \left(\frac{7}{12} n - 0.5\right)(w + s)\right] \tag{7}$$

An asymmetric spiral inductor has two slightly different values of self-resonance frequencies, SRF_1 and SRF_2, when Terminal 1 or Terminal 2 is grounded, respectively. To reproduce such asymmetry in the model, the capacitances C_{P1} and C_{P2} are calculated by expressions (6) and (7), respectively.

The series resistance, R_S, is modeled as in [6]. Its expression is reported below for the sake of clarity:

$$R_{\mathrm{s}} = R_{\mathrm{DC}} \left[1 + \left(\frac{freq}{f_{\mathrm{R0}}} - \frac{freq^2}{2 f_{\mathrm{R0}} f_{\mathrm{RM}}}\right) \left(\frac{f_{\mathrm{R0}}}{2 f_{\mathrm{RM}}} - 1\right)^{-1}\right] \tag{8}$$

where R_{DC} is the overall dc resistance of the inductor, while f_{RM} and f_{R0} are the frequencies for which R_S assumes its maximum and zero, respectively. The values of f_{RM} and f_{R0} are calculated by means of the different empirical formulas

$$f_{R0} \approx (12 \cdot 10^3 w + 1.56) \cdot SRF_1 \quad (9)$$

$$f_{RM} \approx (6.13 \cdot 10^3 w + 0.65) \cdot SRF_1 \quad (10)$$

Finally, substrate losses are modeled by means of R_P according to the following novel expression:

$$R_P = \frac{\rho_{Si} t_{Si}}{\pi \left(\frac{d_{OUT}}{2}\right)^2} \frac{2}{1 + e^{kp * freq}} \quad (11)$$

where ρ_{Si} and t_{Si} are the resistivity and the thickness of the silicon substrate, d_{OUT} is the inductor outer diameter, and k_P is equal to 10^{-20} Hz^{-1}. It is worth noting that the adopted R_P expression better predicts the resistive loss increase with frequency in comparison with the equivalent one reported in [6].

3.2. Advanced Lumped Model

The lumped model shown in Figure 5b is fully geometrically scalable as the first one, but it exploits an advanced topology with thirteen components. Thanks to a more distributed and coupled topology, it better predicts the two-port behavior of spiral inductors, that can hardly be modeled by using a simple π-like network [10].

To this aim, the inductance, L, which is still calculated by means of Equations (1) and (2), is split into three magnetically coupled components (i.e., L_M and two equal inductances, L_S) according to the spitting factor, k_s. The expressions for L_M and L_S are reported below:

$$\frac{L_M}{L} = \frac{1 - 2k_S}{1 + 4k\sqrt{k_S(1 - 2k_S)}} \quad (12)$$

$$\frac{L_S}{L} = \frac{k_S}{1 + 4k\sqrt{k_S(1 - 2k_S)}} \quad (13)$$

The magnetic coupling factor, k, has been set as equal to 0.5, assuming that, regardless of the inductor geometry, just half of the whole magnetic flux generated by $\boldsymbol{L}_M$ is linked to $\boldsymbol{L}_S$. The splitting factor, k_S, has been set to 2.5×10^{-2} to fit the measured S-parameters.

The model exploits expressions (3) and (11) for C_S and R_P, respectively. On the other hand, $C_{P1,2}$ is calculated by means of the improved Equations (14)–(16):

$$C_{P1} = \left(\frac{\beta_2}{w} + \frac{2}{3}\right) C_{SUB} A_{OUT} \quad (14)$$

$$C_{P2} = \frac{2}{3} C_{SUB} A_{OUT} \quad (15)$$

$$A_{OUT} = \pi \left(\frac{d_{OUT}}{2}\right)^2 \quad (16)$$

where C_{SUB} is the substrate capacitance per unit area and d_{OUT} is the external diameter of the spiral. The term (β_2/w) accounts for the asymmetry of the capacitances C_{P1} and C_{P2} due to asymmetrical inductor layout. Moreover, it models the increase in capacitance due to fringing effects, which are relevant for small-width inductors. In this respect, similar considerations about the electric fringing phenomena affecting the substrate capacitances are reported in [6]. The parameter β_2 has been set as equal to 2 μm.

Finally, thanks to inductance splitting and magnetic coupling, the topology allows adopting a physical expression to model the monotone increase in the series resistance, R_S, due to current crowding phenomena [31,37–40], which is not possible in a simple π-like model such as the one shown in Figure 5a. Specifically, R_S is modeled with Equations (17) and (18):

$$R_S = R_{DC}\left[1 + \frac{f}{f_0}\right] \tag{17}$$

$$f_0 = \alpha_2 SRF_1 \tag{18}$$

where R_{DC} is the overall dc resistance of the inductor. The parameter α_2 has been set as equal to 0.47 to model the normalized series resistance, R_S/R_{DC}, as shown in Figure 6. It is worth noting that R_S is on average around three times its DC value for $f = SFR_1$.

Table 1 summarizes the dependencies of performance parameters on circuit components, fitting parameters and scalable equations for the advanced model in Figure 5b. A sensitivity analysis for the main performance parameters is instead reported in Table 2.

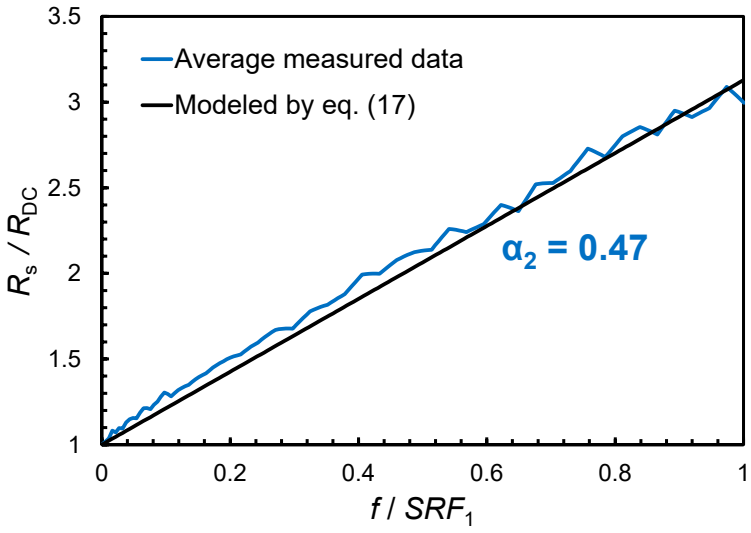


Figure 6. Comparison between measured and modeled normalized series resistance.

Table 1. Performance parameter dependencies for the advanced model in Figure 5b.

Performance Parameters	Lumped Components	Fitting Parameters	Scalable Equations
SRF_1	L, C_{P1}, C_S	k, k_S, β_2	(1), (2), (3), (12), (13), (14)
Q-factor	L, R_S, R_P	α_2, k, k_S, k_P	(1), (2), (12), (13), (17), (18)
$\lvert S_{11}\rvert$	All	k, k_S, k_P	All
$\lvert S_{21}\rvert$	All	α_2, k, k_S, k_P	All

Table 2. Sensitivity analysis on performance parameters for 50% variations in the fitting parameters for the advanced model in Figure 5b (n = 1.5, w = 60 μm, d_{IN} = 290 μm).

Performance Parameters	k	k_S	α_2	β_2	k_P	Unit
ΔSRF_1	2.5	1.7	-	−1.2	-	%
ΔQ_{MAX}	0.7	0.5	15.6	−0.8	−0.6	%
$\Delta\lvert S_{11}\rvert @\ SRF_1$	−0.83	−0.78	0.03	−0.03	−0.17	dB

$\Delta\lvert S_{21}\rvert$@ SRF_1	0.16	0.15	0.04	0.01	−0.15	dB

4. Model Validation and Comparison

The proposed lumped models have been validated by comparison with the experimental data of geometrically scaled circular inductors in a very wide range of geometrical parameters. Specifically, the turn number, n, spans from 1.5 to 6.5, the metal width, w, from 10 μm to 100 μm, and the inner diameters, d_{IN}, from 60 μm to 300 μm, while the metal spacing, s, is fixed to the minimum technological value, i.e., 10 μm. In this work, special attention was given to covering a wide range of metal width, w (i.e., 10 μm to 100 μm), which is essential to achieve accuracy not only for low-current inductors, but also for high-current inductors in power applications. As far as the frequency range is concerned, both models have been developed to cover typical needs of sub 6-GHz power amplifiers and therefore they have not been validated for high-frequency inductors, i.e., inductors with *SRF* higher than 15 GHz, which exhibit a *Q*-factor peak well above 6 GHz.

The comparison is first carried out in terms of the common inductor performance parameters, such as *L*, *Q*-factor, and *SRF*. Indeed, such parameters, along with the maximum delivered current and consumed area, are largely adopted by RF IC designers to optimize the geometry of the inductor, as already shown in Figure 4.

Figures 7–10 compare measured inductance and *Q*-factor curves versus frequency with the estimated behaviors of the two lumped models for four geometrically scaled inductors, whose layouts are shown for the sake of clarity. Figure 11 compares model-calculated values of low-frequency *L*, *SRF*, and *Q*-factor at 2 GHz with measured values. Both models approximate the low-frequency *L* with maximum errors below 8%. On the other hand, both *SRF* and *Q*-factor estimations are very challenging due to the very wide geometrical parameter range. Specifically, as far as the *SRF* is concerned, the π-based model in Figure 5a shows a lack of accuracy at the increasing of the metal width, w, with percentage errors as high as 13%. On the other hand, both models predict the *Q*-factor with good precision in the rising low-frequency region, where the skin effect is dominant, while showing growing errors at higher frequencies when current crowding is significant and resonance bend starts. The *Q*-factor peak frequency range is well predicted, but maximum *Q*-factor is affected by higher errors for both models. Finally, the advanced model in Figure 5b achieves better accuracy in the *SRF* prediction with maximum errors of about 3%.

Discrepancies between models and measurements can be attributed to the substrate effects. Indeed, series losses due to current crowding are well predicted, as demonstrated by Figure 6, and therefore the residual error on the *Q*-factor is mainly ascribed to the substrate losses. It is worth noting that adopted silicon substrate exhibits frequency-dependent conductivity [6], which cannot be accurately modeled by a simple equation like (11) used in both presented models. The frequency dispersity could also explain the inductance discrepancies since the inducted magnetic field, expressed by L_{SUB} in Equation (1), depends on the substrate conductivity.

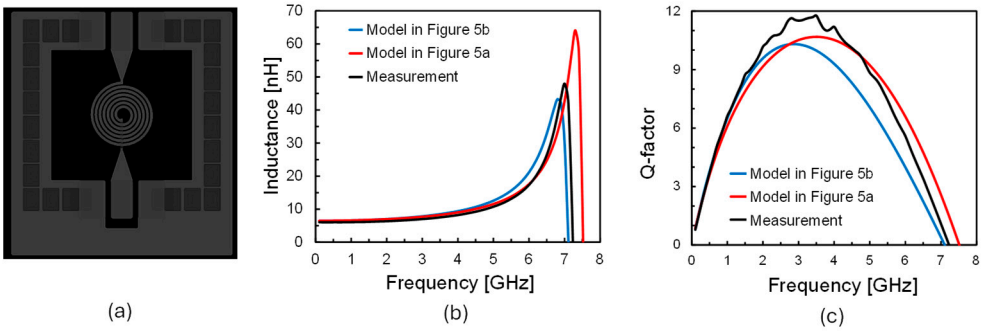


Figure 7. Comparison between proposed models and measurements: (**a**) circular inductor layout (n = 6.5, w = 10 μm, d_{IN} = 60 μm), (**b**) inductance versus frequency, (**c**) Q-factor versus frequency.

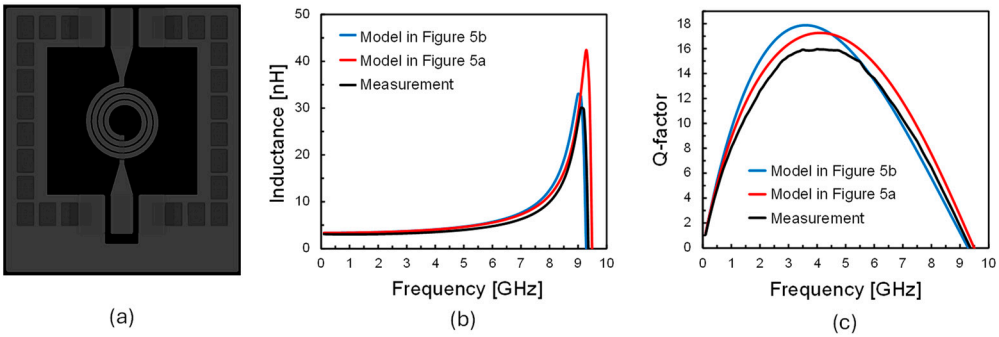


Figure 8. Comparison between proposed models and measurements: (**a**) circular inductor layout (n = 3.5, w = 20 μm, d_{IN} = 150 μm), (**b**) inductance versus frequency, (**c**) Q-factor versus frequency.

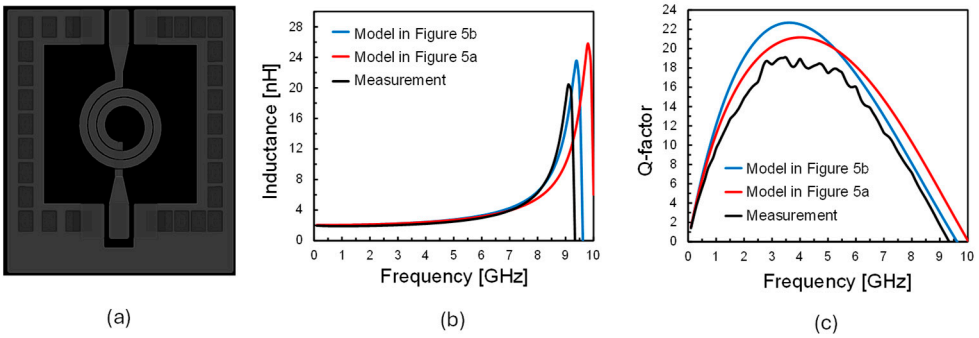


Figure 9. Comparison between proposed models and measurements: (**a**) circular inductor layout (n = 2.5, w = 40 μm, d_{IN} = 200 μm), (**b**) inductance versus frequency, (**c**) Q-factor versus frequency.

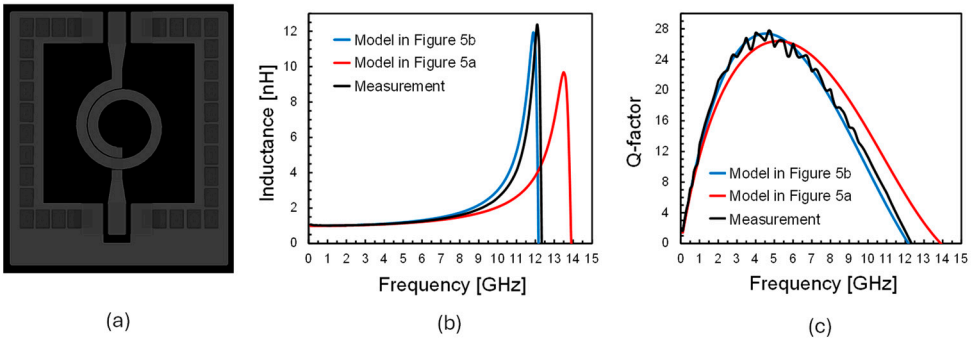


Figure 10. Comparison between proposed models and measurements: (**a**) circular inductor layout (n = 1.5, w = 60 μm, d_{IN} = 290 μm), (**b**) inductance versus frequency, (**c**) Q-factor versus frequency.

The comparison between models has also been carried out in terms of two-port *S*-parameters in Figures 12–15. Indeed, accurate *S*-parameter estimation is required to exploit lumped modeling in RF IC design, when inductor EM-extracted data are not available yet. Figures 11–14 compare measured and modeled S-parameters (i.e., S_{11}, S_{22}, and S_{21}) for the previously reported geometries. The comparison highlights that the *S*-parameter estimation of the π-model has significant discrepancies with respect to measurements, especially for higher *w* values. Moreover, the π-model shows its limits in predicting the transmission coefficient, S_{21}, which is of high importance to evaluate inductor signal loss. On the other hand, Figures 10–14 demonstrate that the advanced model in Figure 5b is well-suited to *S*-parameter prediction thanks to its topology.

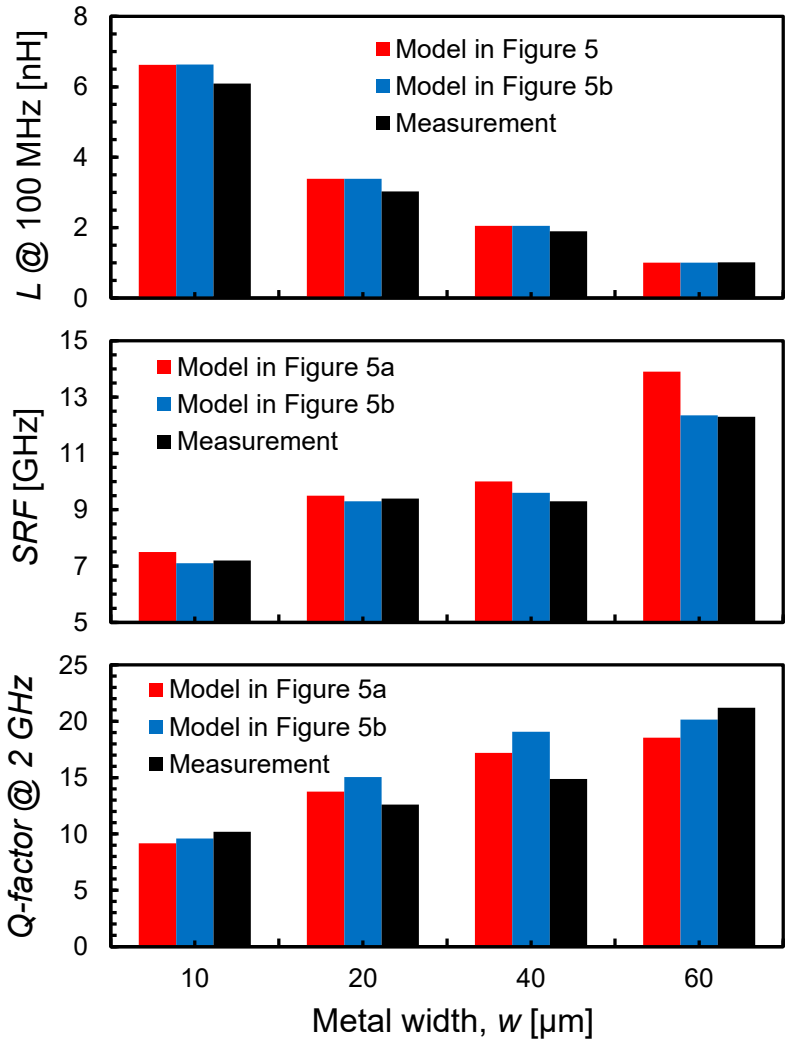


Figure 11. Model comparison at different metal widths, *w*.

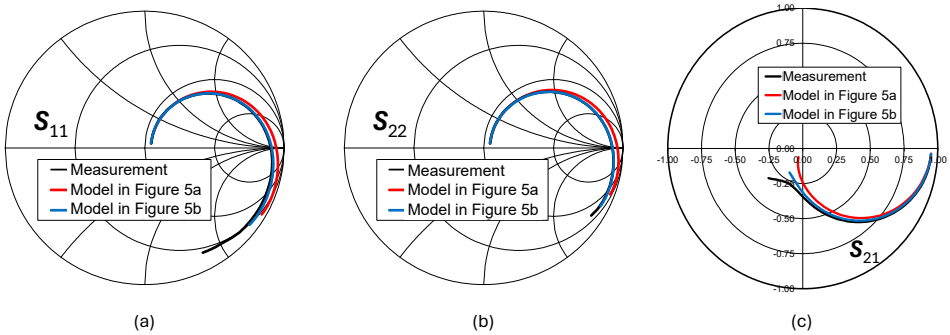


Figure 12. Comparison between model-estimated and measured *S*-parameters: (**a**) S_{11}, (**b**) S_{22}, and (**c**) S_{21} (n = 6.5, w = 10 μm, d_{IN} = 60 μm).

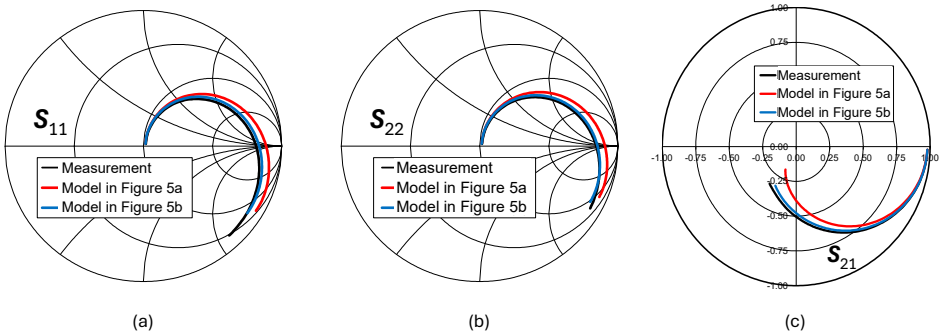


Figure 13. Comparison between model-estimated and measured *S*-parameters: (**a**) S_{11}, (**b**) S_{22}, and (**c**) S_{21} (n = 3.5, w = 20 μm, d_{IN} = 150 μm).

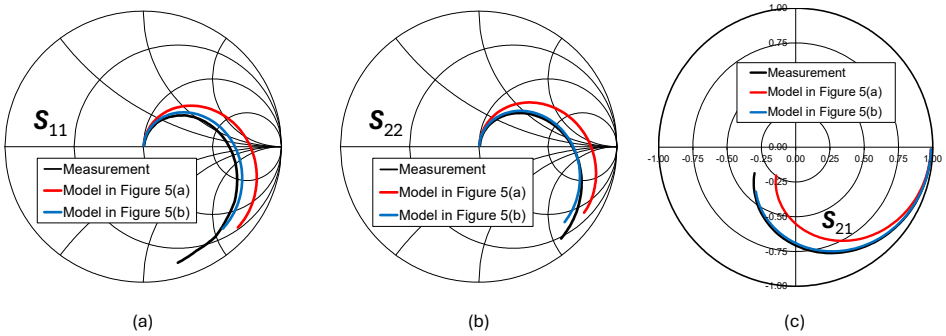


Figure 14. Comparison between model-estimated and measured *S*-parameters: (**a**) S_{11}, (**b**) S_{22}, and (**c**) S_{21} (n = 2.5, w = 40 μm, d_{IN} = 200 μm).

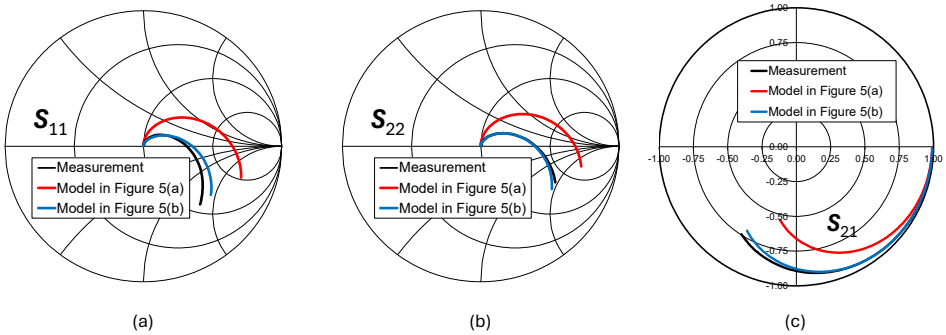


Figure 15. Comparison between model-estimated and measured *S*-parameters: (**a**) S_{11}, (**b**) S_{22}, and (**c**) S_{21} (n = 1.5, w = 60 μm, d_{IN} = 290 μm).

Specifically, the transmission coefficient, S_{21}, is predicted with lower percentage errors compared to the simple π-model ones, as shown in Figure 16 for four scaled inductors.

The above results confirm the soundness of both lumped scalable models for GaN inductors that can be profitably used within the design flow in Figure 4 (steps 1–2), while the advanced model is more suited for the tuning phase (step 3) that requires accurate S-parameter estimation. To the author's knowledge, this is the first time a fully scalable model has been experimentally validated for *S*-parameter prediction of GaN-on-Si inductors.

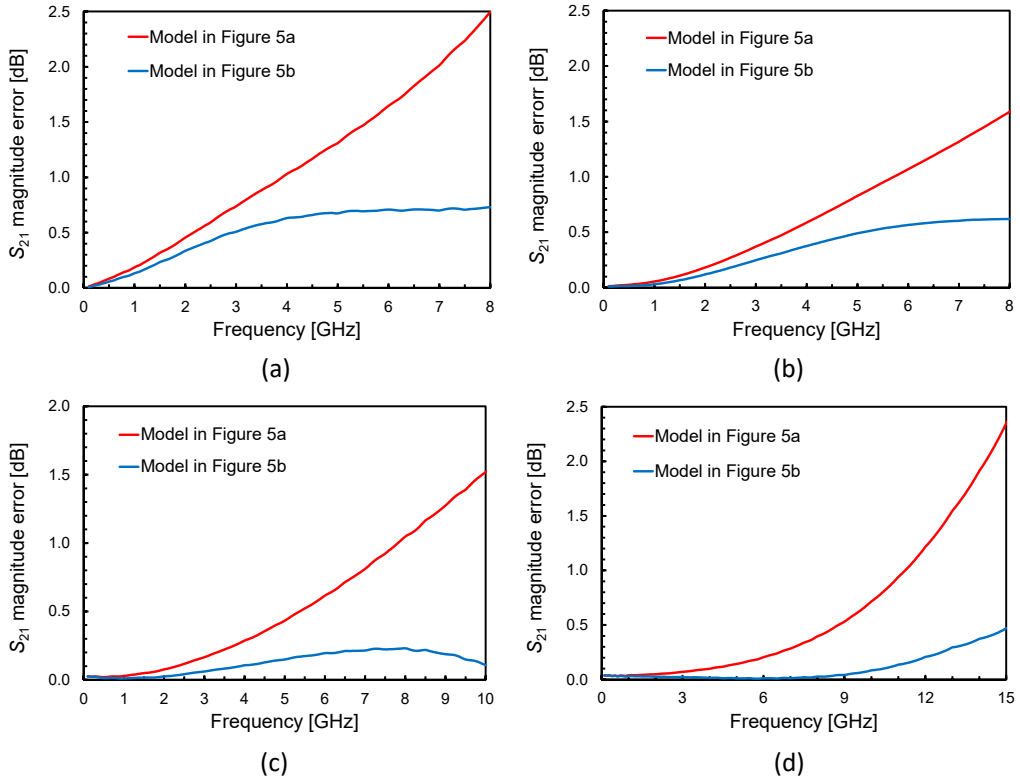


Figure 16. S_{21} magnitude errors**:** (**a**) n = 6.5, w = 10 μm**,** d_{IN} = 60 μm, (**b**) n = 3.5**,** w = 20 μm, d_{IN} = 150 μm, and (**c**) n = 2.5**,** w = 40 μm, d_{IN} = 200 μm, (**d**) n = 1.5, w = 60 μm, d_{IN} = 290 μm.

5. Conclusions

This study proposes and compares a π-based model and an advanced model for circular inductors in 0.5 μm RF GaN-on-Si technology for sub 6-GHz power applications. Both models are lumped and fully geometrically scalable. They demonstrate similar accuracy in predicting both *L* and *Q*-factor in a wide range of geometrical parameters, also including high-width values adopted for power applications. Moreover, thanks to its novel topology, the advanced model achieves improved accuracy in *SRF*, as well as in *S*-parameter estimation, which makes it highly suited to typical RF IC design flow. To the authors' knowledge, this is the very first experimentally validated model for spiral inductors in RF GaN technology able to predict two-port *S*-parameters in a wide range of geometrical parameters. The high potentiality of the model suggests that in future work it could be extended either to a wider range of geometric parameters and therefore higher operative frequencies or for different shapes or structures (i.e., polygonal, multi-layer, symmetric, etc.). Finally, the model could also be improved by taking advantage of AI-based techniques that have already proven useful in this field.

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Data Availability Statement: The data presented in this study is available on request from the corresponding author.

Conflicts of Interest: Simone Spataro, Giuseppina Sapone, and Marcello Giuffrida are employed by the company STMicroelectronics. The remaining author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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